SJSU Hackathon: FlashAttention Inspired Tiled Causal Self Attention Accelerator

Team Name: Team Fafada Jalebi Chavana

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Problem Domain:

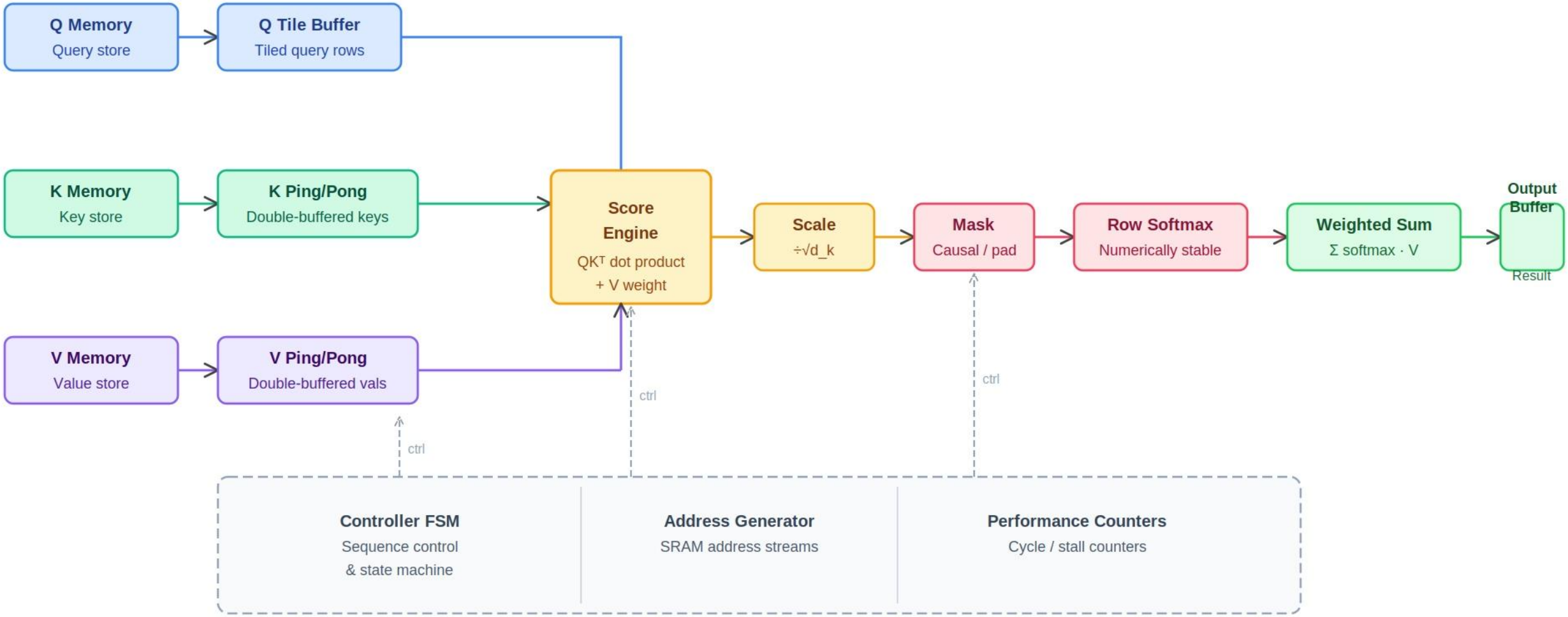
This project is in the domain of AI hardware acceleration, specifically transformer inference hardware. Modern transformer models rely heavily on the self-attention operation, which becomes expensive in both computation and memory as sequence length increases. A major bottleneck is the generation and storage of the attention score matrix, which grows quadratically with sequence length. This creates heavy memory traffic and limits performance and efficiency in hardware implementations. We selected this domain because transformer acceleration is one of the most relevant problems in modern digital design, and it combines architectural thinking, RTL design, memory scheduling, and numeric approximation in one project. Our background is in RTL design, digital systems, FPGA-based implementation, verification, and SystemVerilog-based hardware development. This makes transformer attention acceleration a strong fit because it allows us to apply practical hardware design skills to an important AI workload.

Proposed Innovation:

1) Our project proposes a FlashAttention-inspired tiled causal self-attention accelerator written in SystemVerilog. The main idea is to avoid storing the full attention score matrix in external memory. Instead, the design processes attention in small tiles, reuses on-chip buffers, and performs row-wise softmax computation in hardware. This reduces memory movement and makes the architecture more efficient and more realistic for accelerator-oriented RTL design.

2) The architecture contains a query tile buffer, ping-pong key/value tile buffers, a score engine for tiled dot-product computation, a causal mask unit, a LUT-based softmax pipeline, a weighted-sum engine, an output buffer, and a controller FSM. The ping-pong buffering scheme allows one key/value tile to be used in computation while the next tile is loaded, improving utilization and making the design more ambitious than a basic attention implementation.

3) We will apply AI in a practical engineering workflow. AI will help generate RTL skeletons, testbench scaffolding, lookup-table contents for exponential approximation, block diagrams, and software reference models for output checking. AI will also accelerate debugging by helping isolate interface mismatches, FSM issues, and fixed-point errors.

4) We will test the final product using a SystemVerilog testbench. Demonstration will include correct attention outputs for fixed test cases, causal masking behavior, tile-by-tile execution waveforms, and performance counters showing cycles, tile reuse, and reduced score-storage requirement.

Attention Hardware Accelerator — Datapath